GEORGE JAMESON OBED

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SUMMARY PROFILE

Biologist and microbial ecologist completing an MSc in Natural Resources Management – Biology at the Norwegian University of Science and Technology (NTNU). I have interdisciplinary experience in microbiology, microbial ecology, molecular biology and environmental research, most of it carried out in the semi-arid environments of central Tanzania. My work spans soil-associated and root-nodule bacteria, nitrogen fixation, antimicrobial resistance in wildlife from a One Health perspective, and bioprospecting for plant-derived antimicrobials, combining ecological field research, culture-based microbiology and DNA-based methods. I am particularly interested in how microbial diversity and function respond to environmental conditions, soil properties and human activities, especially in drylands and other extreme environments, and in how this knowledge can inform the sustainable management of natural resources. Alongside research, I have trained smallholder farmers in soil health and taught undergraduate microbiology and parasitology.

EDUCATION BACKGROUND

Aug 2025 – May 2027 **MSc Natural Resources Management – Biology**
Norwegian University of Science and Technology (NTNU), Trondheim, Norway
Thesis Title: Ecological and Anthropogenic Correlates of Zoonotic and Antimicrobial-Resistant Bacteria in Invasive Indian House Crows (*Corvus splendens*) in Dodoma Urban, Tanzania
Relevant areas: *Microbial ecology, Wildlife diseases, Antimicrobial resistance, Human-wildlife interaction & One Health*

Nov 2019 – Dec 2022 **BSc Biology**
University of Dodoma (UDOM), Dodoma, Tanzania
Research Project Title: The Antimicrobial Activities of *Moringa oleifera* Seed and Leaf Extracts Against Selected Pathogenic Microbes
Relevant areas: Microbiology, Bioprospecting, Antimicrobial resistance, Drug discovery & Traditional medicine.

RESEARCH EXPERIENCE

Jul 2026 – Apr 2027 **MSc Researcher**
Norwegian University of Science and Technology (NTNU), Dodoma CC, Dodoma, Tanzania
Project Title: Ecological and Anthropogenic Correlates of Zoonotic and Antimicrobial-Resistant Bacteria in Invasive Indian House Crows (*Corvus splendens*) in Dodoma Urban, Tanzania
Supervisors: Prof. Augustine Arukwe, Prof. Eivin Røskaft, Dr. Peter Ranke & Dr. Ignas Safari.

- Captured 53 Indian House Crows using baited ladder traps and collected gut contents for bacterial isolation and antimicrobial resistance testing.
- Collected environmental and anthropogenic data associated with bacterial carriage and antimicrobial resistance.
- Isolated and identified enteric bacteria using culture and biochemical tests and assessed antimicrobial susceptibility using the Kirby–Bauer disk diffusion method.
- Analyzing ecological, environmental, and microbiological data to evaluate associations among sampling sites, anthropogenic factors, bacterial carriage, and antimicrobial resistance patterns.

Jul 2023 – Present **Research Assistant & Project Team Member**
Pennsylvania State University (PSU), University of Dodoma (UDOM) & Community Development and Empowerment Organization (CDEO), Dodoma, Tanzania
Project Name: Project to Improve Soil Health and Fertility in Nghumbi (MKARUN)
Principal Investigator: Kelsey Christine Mercurio

- Sampled root nodules from drought-tolerant legumes on smallholder farms in a semi-arid region across two seasons.
- Extracted DNA from root nodules for downstream 16S and nifH amplicon sequencing and analysis.
- Developed a culture collection of nitrogen-fixing bacteria and other endophytes from root nodules using semi-selective and non-selective media.
- Provided research design and methodological advice on Gliricidia-based agroforestry and cover crop–microbiome studies, supporting study planning, sampling approaches, and research methods.

Nov 2023 **Research Assistant**
Society for the Protection of Underground Networks (SPUN) & University of Dodoma, Dodoma, Tanzania
Project Title: Molecular diversity of Arbuscular Mycorrhizal fungi in different land uses in Dodoma, Tanzania.
Principal Investigator: Joseph Innocent Massawe

- Executed field sampling protocols for soil microbiomes and conducted mycorrhizal DNA extractions for molecular diversity mapping across varied land-use gradients.

Jan 2022 – May 2022 **BSc Research Project**
University of Dodoma (UDOM), Dodoma, Tanzania
Project Title: The Antimicrobial Activities of *Moringa oleifera* Seed and Leaf Extracts Against Selected Pathogenic Microbes
Supervisor: Dr. Mhuji Kilonzo

- Prepared crude extracts from *Moringa oleifera* leaves and seeds using the maceration method.
- Prepared different concentrations of *Moringa oleifera* leaf and seed crude extracts and evaluated their antimicrobial activity using the agar well diffusion method against *Escherichia coli, Salmonella* Typhi, and *Candida albicans*.
- Documented and analyzed the experimental findings and prepared a research report as part of the requirements for completion of the BSc Biology degree.

TEACHING, EXTENSION AND OUTREACH

Jul 2023 – Present **Research Assistant & Project Team Member**
Pennsylvania State University (PSU), University of Dodoma (UDOM) & Community Development and Empowerment Organization (CDEO), Dodoma, Tanzania
Project Name: Project to Improve Soil Health and Fertility in Nghumbi (MKARUN)
Principal Investigator: Kelsey Christine Mercurio

- Implemented training in soil health concepts for smallholder farmers in a semi-arid context.
- Facilitated group decision-making for a multi-farm cover crops field trial.
- Co-developed farmer monitoring tools for integrated pest management, including identification of local insect species, establishment of user-friendly sampling protocols, and translation from English to Swahili.

Dec 2022 – Jan 2025 **Volunteer Laboratory Scientist – Biology**
University of Dodoma (UDOM), College of Natural and Mathematical Sciences (CNMS), Department of Biology, Dodoma, Tanzania

- Taught undergraduate students about theoretical concepts in microbiology and parasitology before lab work.
- Supervised undergraduate students during microbiology and parasitology laboratory practicals.
- Assisted with the management and maintenance of the Biology department’s apiary project.

INTERNSHIP EXPERIENCE

Aug 2021 – Oct 2021 **Internship on Agroforestry and Seed Development and Multiplication**
Tanzania Agricultural Research Institute (TARI), Tumbi Centre, Tabora, Tanzania

- Propagated and managed indigenous and fruit-tree species through hands-on seed development and multiplication work, including *Vitex mombasae*, *Vitex doniana*, *Strychnos cocculoides*, marula (*Sclerocarya birrea*), cashew (*Anacardium occidentale*), citrus, and mango.

TRAINING AND PROFESSIONAL DEVELOPMENT

Jul 2024 **CIRTification: Community Involvement in Research Training – Human Research Protections Training**
University of Illinois Chicago, Center for Clinical and Translational Science, Online

- Completed training covering research ethics, participant eligibility and recruitment, informed consent, Institutional Review Boards, data collection and protection, and ethical field research practices.

Mar – Apr 2023 **One Health Pathogen Genomics and Metagenomics — Hands-on Training**
Southern Africa Center for Infectious Diseases Surveillance (SACIDS), Government of the United Republic of Tanzania & Sokoine University of Agriculture (SUA)

- Received hands-on training in DNA extraction, agarose gel electrophoresis, and DNA quality and concentration assessment using Qubit and NanoDrop.
- Gained practical experience with Oxford Nanopore sequencing technology and sequence data processing.
- Performed sequence similarity and identification analyses using BLAST (Basic Local Alignment Search Tool).

RESEARCH METHOD AND TECHNICAL SKILLS

Microbiology
Sterilization techniques
Bacterial and fungal culture, isolation and identification
Biochemical identification of bacteria (TSI, Indole, Motility, H_2S, Citrate, Methyl red, Urease tests)
Gram staining
Antimicrobial susceptibility testing (Kirby–Bauer Disk Diffusion Method)

Ecological / Field Methods
Ecological and environmental data collection
Collection and processing of diverse sample types for microbiome analysis, including soils, plants, and animals
Bird capturing
Species identification

Molecular Biology
DNA extraction
DNA/RNA sample handling and storage
Agarose gel electrophoresis
DNA quantification and quality assessment — Qubit, NanoDrop
PCR
Oxford Nanopore sequencing

Research and Data analysis Skills
Research study design
Literature review
Scientific documentation (proposal and report writing)
BLAST sequence analysis
Data cleaning and visualization, and analysis of 16S OTU tables in R
Linear models and Discriminant analysis in R
Zotero

SCHOLARSHIPS AND AWARDS

Aug 2025 – May 2027 **NORSTIP Scholarship**
Norwegian Directorate for Higher Education and Skills

PROFESSIONAL AFFILIATIONS

Nov 2025 – Present **Member**
British Ecological Society (BES)

Jan 2026 – Present **Member**
American Ornithological Society (AOS)

PUBLICATIONS, REPORTS, & PRESENTATIONS

Reports:

Røsholt-Høgslund, A., Loya Loya, B., **Obed, G. J.**, Helgetun, H., Teigen, I. M., & Reiersen, M. H. (2025). Implementing OECMs in Norway: Case studies in Trondheim Municipality [Research report]. Norwegian University of Science and Technology. https://doi.org/10.21400/wy6k-5y88

Presentations:

Mercurio, K.C., T.A. Nassari, J.I. Massawe, A.W. Galahenga, S.A. Zglobicki, **G. J. Obed**, M. Msandai, J. Ngumba, J. Missanga, L. Malekela, M. Paulo, M. Kusenha, B. Chimaisi, D. Lemnje, E. Makau, A. Paulo, G. Kaaya, F. Komanya, K.M. Clouse, E. Burkhart, L.T. Burghardt, and the farmer participants of the Mradi wa Kuboresha Afya na Rutuba ya Udongo Nghumbi (MKARUN). Investigating cowpea cover crops and root nodule microbiomes via farmer collaborative research in semi-arid Nghumbi, Tanzania. The 27th North American Symbiotic Nitrogen Fixation Conference, Queens University, Kingston, Toronto, Canada, June 1-4, 2026. Oral and Poster presentations.

Mercurio, K.C., **G.J. Obed**, A.W. Galahenga, G. Kaaya, J. Ngumba, J. I. Massawe, J.S. Missanga, S.A. Zglobicki, T.A. Nassari, M. Msandai, S. Guha, M.-A. Gil-Polo, L.T. Burghardt, and the Farmer Participants of the Mradi wa Kuboresha Afya na Rutuba ya Udongo Nghumbi (MKARUN). Supporting Community-Driven Sustainable Agriculture in Tanzania Using Legume Cover Crops and Participatory Science. Water Conference, Pennsylvania State University, University Park, PA, USA, March 26, 2026. Oral flash talk and Poster presentation.

Manuscripts in Preparation:

Ecological and Anthropogenic Correlates of Zoonotic and Antimicrobial-Resistant Bacteria in Invasive Indian House Crows (*Corvus splendens*) in Dodoma Urban, Tanzania.

Host or Habitat? Discriminant Analysis of Endosymbionts and Other Bacteria as Markers of Sex, Species, Site and Season in Wild Scuttle Flies (Diptera: Phoridae)

LANGUAGES

English – Fluent, Swahili – Native

REFERENCES

Augustine Arukwe	Professor – NTNU, Trondheim, Norway
Eivin Røskaft	Professor Emeritus – NTNU, Trondheim, Norway
Joseph Innocent Massawe	Senior Lecturer – University of Dodoma, Dodoma, Tanzania

Contact details of referees are available on request.